

Foundation of Blockchain Technology (IS501) Pre-Reading Material – Lecture 9

Blockchain Consensus Challenges & Solution

Estimated Reading Time: 15–20 minutes

Why This Topic Matters

A blockchain is maintained by multiple distributed nodes without necessarily having a central authority. A **consensus mechanism** allows these nodes to agree on the valid state of the blockchain and helps maintain consistency and security across the network.

Learning Outcomes

After completing this pre-reading, you should be able to:

- Explain the purpose of a **blockchain consensus mechanism**.
- Understand major consensus mechanisms such as **PoW, PoS, DPoS, and BFT**.
- Identify major challenges associated with blockchain consensus.
- Explain the **51% attack** and its implications.
- Understand selected solutions for blockchain scalability and performance.

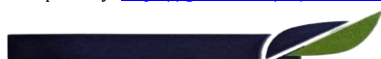
Activate Your Prior Knowledge

1. Distributed Agreement

Suppose thousands of computers maintain the same blockchain.
Think: How can they agree on which transactions should be added without a central authority?

2. Malicious Participants

Imagine some nodes intentionally provide incorrect information.
Think: What mechanism can allow honest nodes to maintain a consistent blockchain?



3. Speed vs Security

A blockchain may become more secure by requiring more computation or verification.

Think: Could increasing security create problems such as higher energy consumption or slower transactions?

Prerequisite Concepts

You should be familiar with:

- Blockchain architecture
- Distributed ledger
- Blockchain nodes
- Public and private blockchain
- Proof of Work and Proof of Stake
- Byzantine Fault Tolerance

Core Concepts

1. Consensus Mechanism

A consensus mechanism is a protocol that enables distributed nodes to agree on a common and valid state of the blockchain. Different mechanisms make different trade-offs between **security, decentralization, speed, and efficiency**.

2. Proof of Work (PoW)

PoW requires nodes called **miners** to solve computational puzzles before adding new blocks. The puzzle is difficult to solve but easy for other nodes to verify.



3. PoW: Advantages and Limitations

PoW can support a large number of participants and provides strong security through computational work. However, solving puzzles through brute force requires substantial computational resources and energy.

4. Proof of Stake (PoS)

PoS replaces computational competition with **economic stake**. Participants become validators based on their stake and help validate transactions and blocks without requiring the same level of computational work as PoW.

5. Delegated Proof of Stake (DPoS)

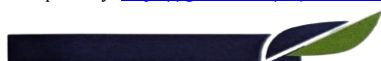
DPoS is a variation of PoS in which stakeholders **vote to elect witness or validator nodes** responsible for creating blocks.

6. Byzantine Fault Tolerance (BFT)

BFT enables a blockchain network to reach consensus even when some nodes fail or behave maliciously. The objective is to maintain consistency among honest nodes despite Byzantine faults.

7. Practical Byzantine Fault Tolerance (PBFT)

PBFT is a BFT-based consensus approach designed to allow distributed systems to continue operating correctly despite Byzantine failures. It uses communication and replication among participating nodes to reach agreement.



8. 51% Attack

A **51% attack** occurs when an attacker gains control of a majority of the relevant network resource or nodes, allowing them to potentially manipulate consensus and fraudulent transactions.

9. Consensus Challenges

Blockchain consensus can face challenges involving **high energy consumption, latency, low throughput, scalability, storage requirements, and security**. Different consensus mechanisms attempt to balance these competing requirements.

10. Scalability Problem

As blockchain transactions increase, the blockchain can experience increased storage requirements, computational demands, latency, and throughput limitations. These issues can make it difficult for all nodes to efficiently maintain and synchronize the blockchain.

11. Scalability Solutions

Several approaches have been proposed to improve blockchain scalability, including **on-chain scaling, off-chain scaling, sidechains, child-chains, and inter-chain solutions**.

12. Off-Chain Solutions

Off-chain approaches move some transactions or processing away from the main blockchain. For example, the **Lightning Network** uses off-chain payment channels to enable faster transactions while reducing the load on the main blockchain.

13. Hybrid Consensus

Hybrid consensus combines more than one consensus mechanism to improve security and resilience. For example, hybrid PoW–PoS systems can require an attacker to control both mining power and stake, increasing the cost of a majority attack.

14. Choosing a Consensus Mechanism

There is no single consensus mechanism that is best for every blockchain. The appropriate mechanism depends on requirements such as **security, decentralization, scalability, transaction speed, energy efficiency, governance, and network participation.**

Guided Reading Questions

1. What is the purpose of a blockchain consensus mechanism?
2. How does PoW achieve consensus?
3. Why does PoW consume significant energy?
4. How does PoS differ from PoW?
5. What is the purpose of DPoS?
6. What problem does BFT address?
7. What is a 51% attack?
8. Why is blockchain scalability a challenge?
9. What are on-chain and off-chain scaling?
10. Why might a blockchain use a hybrid consensus mechanism?

Reflection Activity

Write 100–150 words explaining why there is **no single perfect consensus mechanism** for every blockchain. Consider the trade-offs among **security, decentralization, scalability, transaction speed, and energy consumption.**

Self-Assessment

1. Consensus mechanisms allow blockchain nodes to reach _____.
2. In PoW, participants who solve computational puzzles are called _____.
3. PoW requires significant _____ power.
4. PoS uses economic _____ instead of computational competition.
5. DPoS uses stakeholder _____ to select validators/witnesses.
6. BFT stands for _____.
7. PBFT stands for Practical _____ Fault Tolerance.
8. An attack involving control of a majority of network resources is called a _____ attack.
9. Lightning Network is an example of an _____ scaling solution.
10. Combining multiple consensus mechanisms is called _____ consensus.

Key Takeaway

A blockchain consensus mechanism enables distributed participants to **agree on a common blockchain state without relying on a central authority**. PoW, PoS, DPoS, and BFT provide different approaches, while solutions such as **off-chain scaling and hybrid consensus** attempt to address challenges involving security, scalability, performance, and efficiency.

